

Supply, demand and policy in New York's ride-hail market

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Repository: <https://github.com/caie007/savannahgishon>

Where does supply fail to match demand,
and how do policy and trip economics
relate to market outcomes?



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Data and distributed MapReduce pipeline

MAP

Select fields; derive time, wait and CBD indicators

SHUFFLE

Exchange hashpartitioning groups shared keys

REDUCE

Partial and final sums, counts and means

24 Parquet files • 461.9M cleaned trips • 74.0M aggregate rows

Separate 2025 demand-economics stream: 243.6M completed trips



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Spark execution: measurable distributed work

849 tasks

Full processing run

120.5 GiB

Input read; 7.2 GiB shuffled

12.7 s · RDD

Same aggregation benchmark

1.9 s · DataFrame

Optimised execution

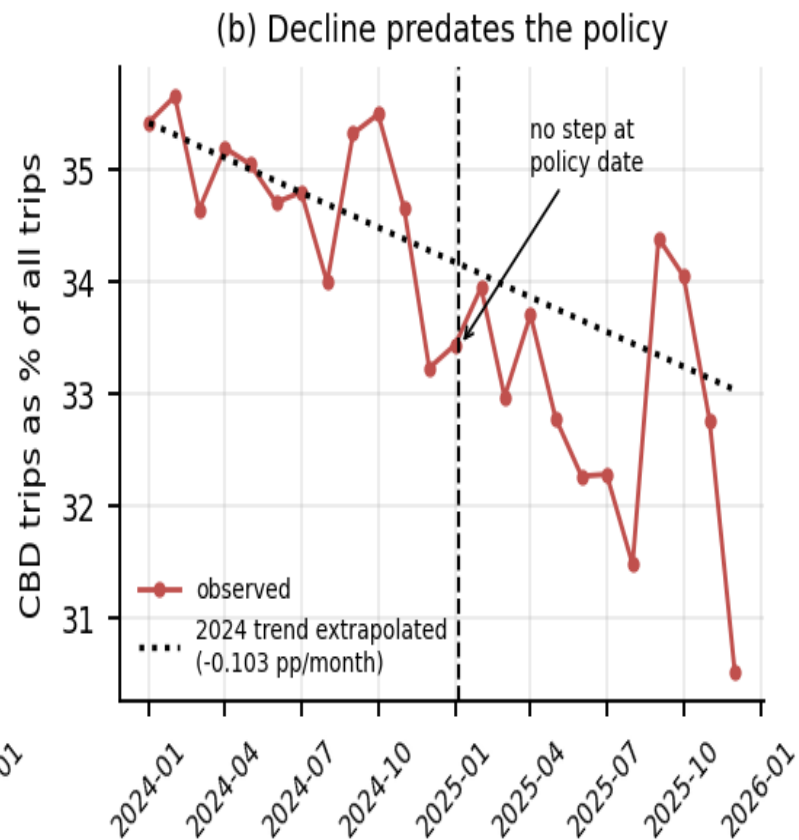
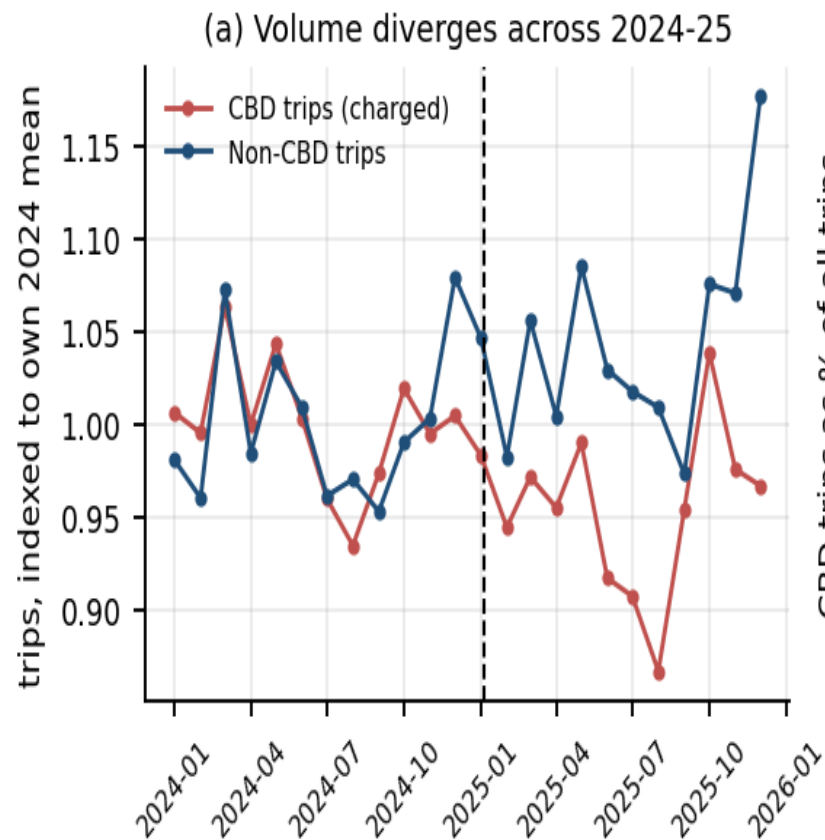


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CBD trips fell, but the decline predates the fee



-4.39%

CBD trips

+4.40%

Non-CBD trips

Pre-policy CBD share was already falling.

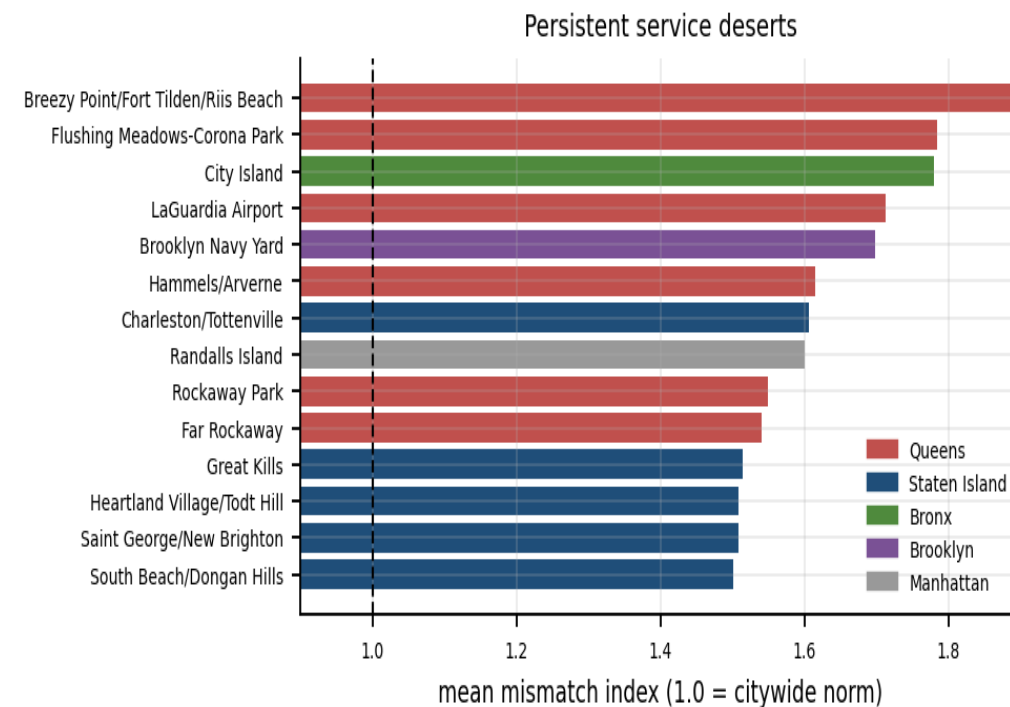
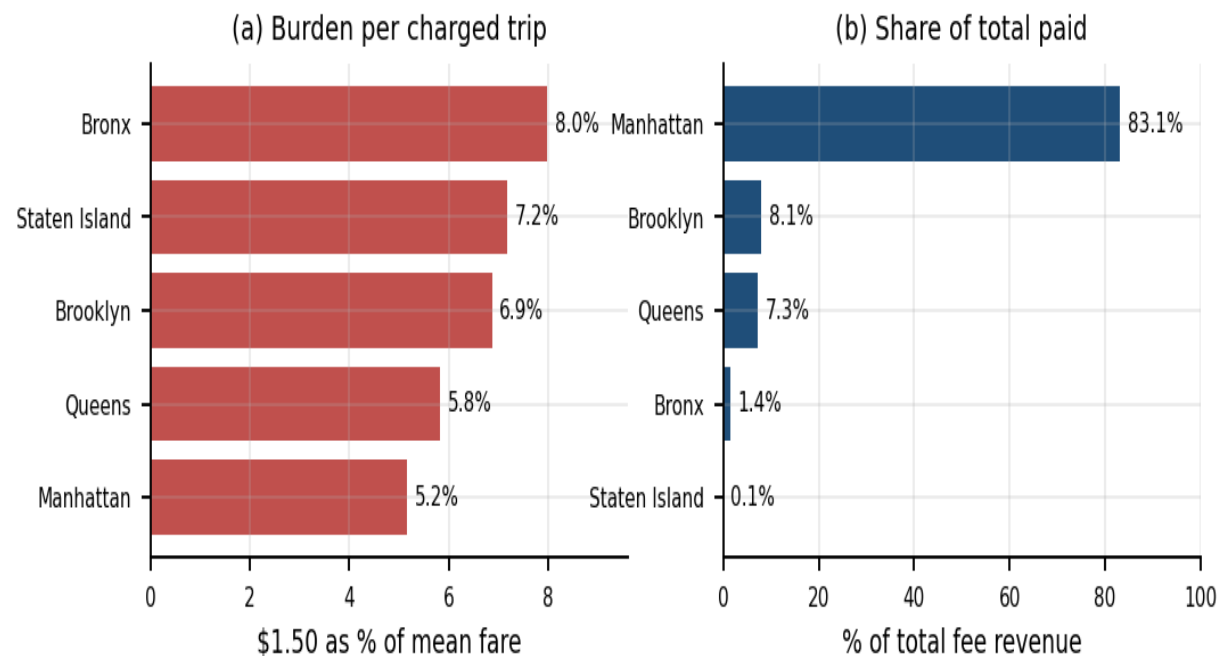


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Who paid, and who waited?



Fee burden: Bronx 7.98% vs Manhattan 5.17%

74 persistent service-desert zones



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How we classified realised demand

243.6M completed trips → 8,760 date-hour counts

LOW

$\leq 25,205$ trips/hour

MEDIUM

25,206–33,474 trips/hour

HIGH

$> 33,474$ trips/hour

Broadcast hourly categories back to individual trips

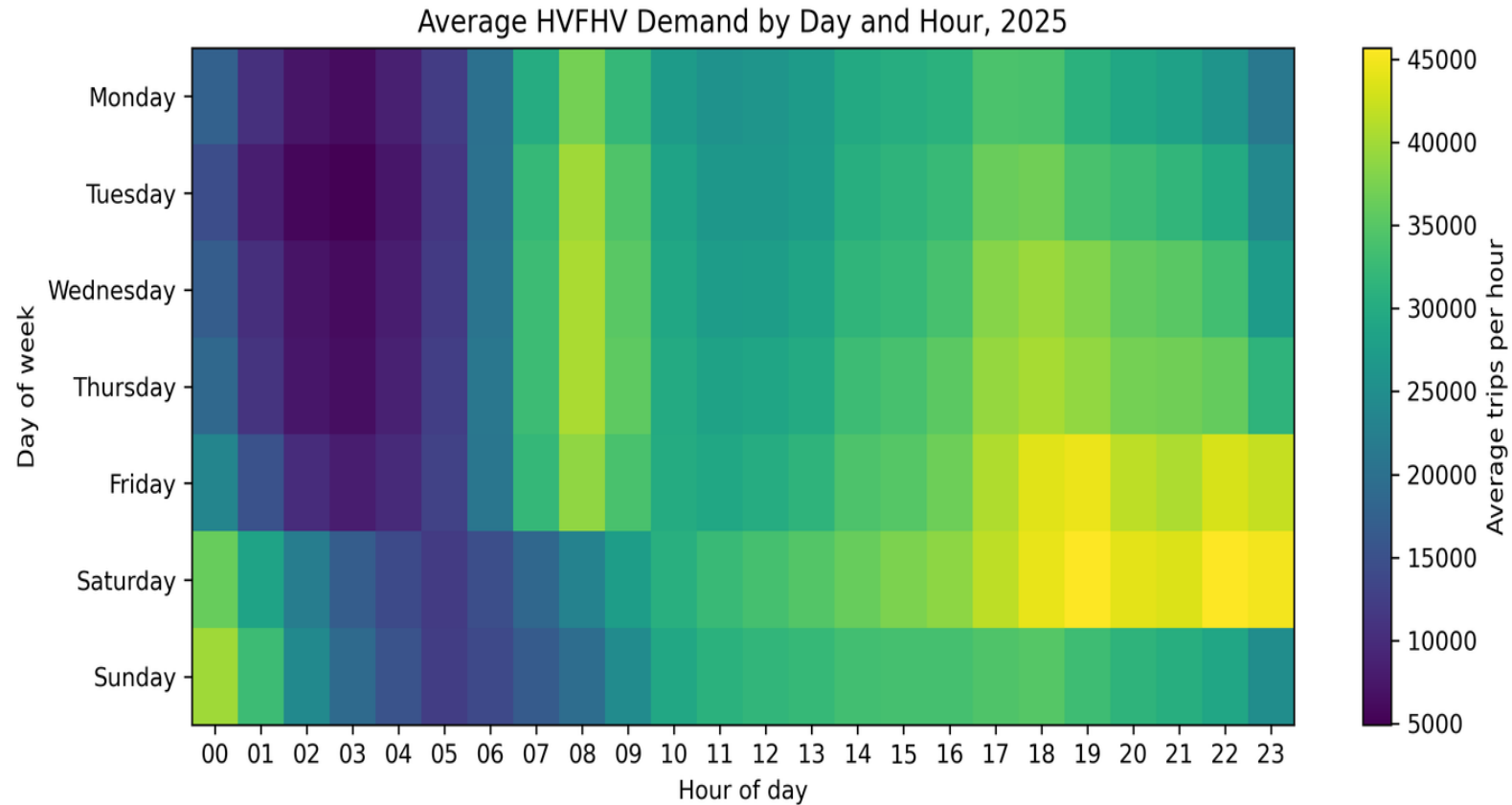


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Demand varies by hour and day



03:00 $\approx$ 9,699/hour | 18:00 $\approx$ 39,127/hour

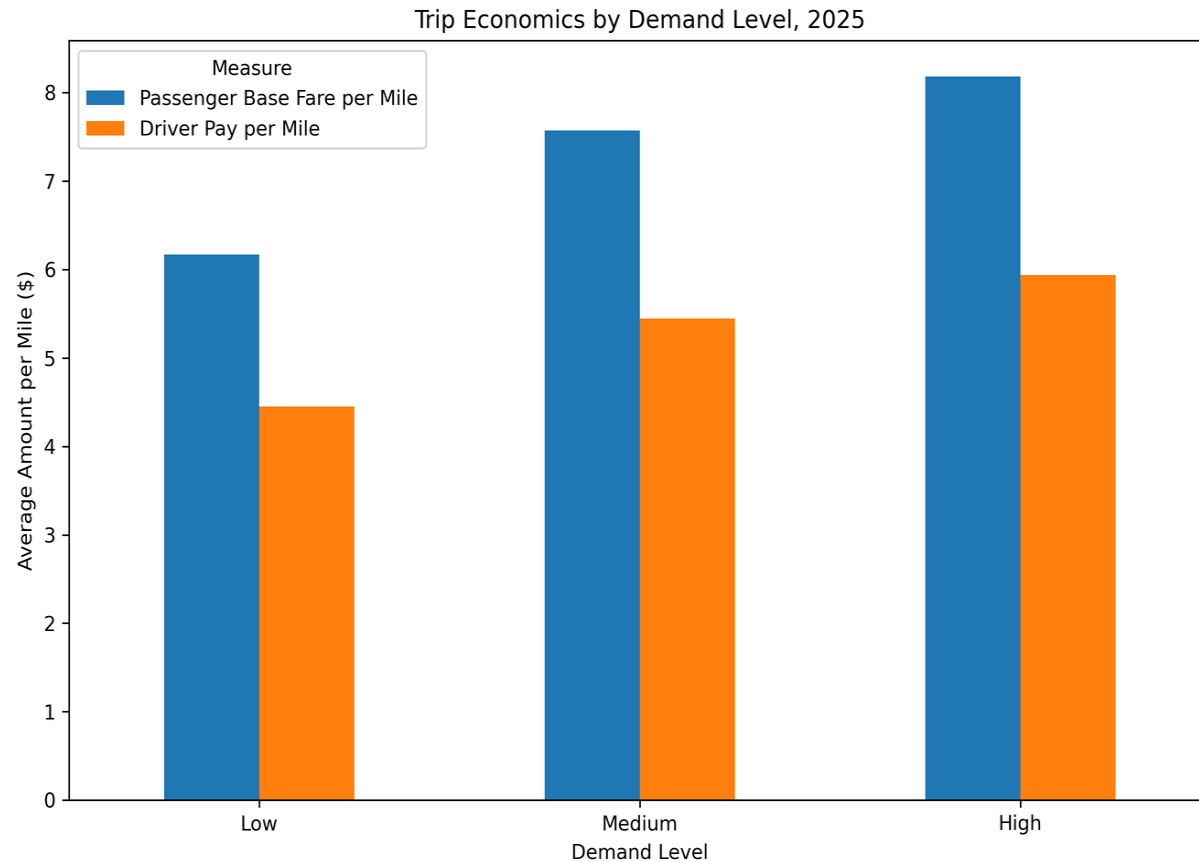


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Trip economics: per-mile measures rise



Fare per mile

\$6.17 → \$8.18

Driver pay per mile

\$4.45 → \$5.94

Distance: 6.00 → 4.68 mi | Duration: 17.85 → 20.43 min



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Demand analysis: PySpark code workflow

1 · **TRANSFORM** pickup_date, hour, trip_minutes

2 · **AGGREGATE** groupBy(date, hour).count()

3 · **BROADCAST JOIN** Attach Low / Medium / High to trips

4 · **REDUCE** Mean fare, driver pay, distance and duration



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Practical value and limitations

OPERATORS

Plan driver availability by time and place

DRIVERS

Higher pay per mile does not equal profit

PASSENGERS

Similar total fares can mean different cost per mile

LIMITATIONS

Completed trips $\neq$ total requests; association $\neq$ causation



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